ASA International Data Quest Türkiye 2026

Low fertility, continued growth?

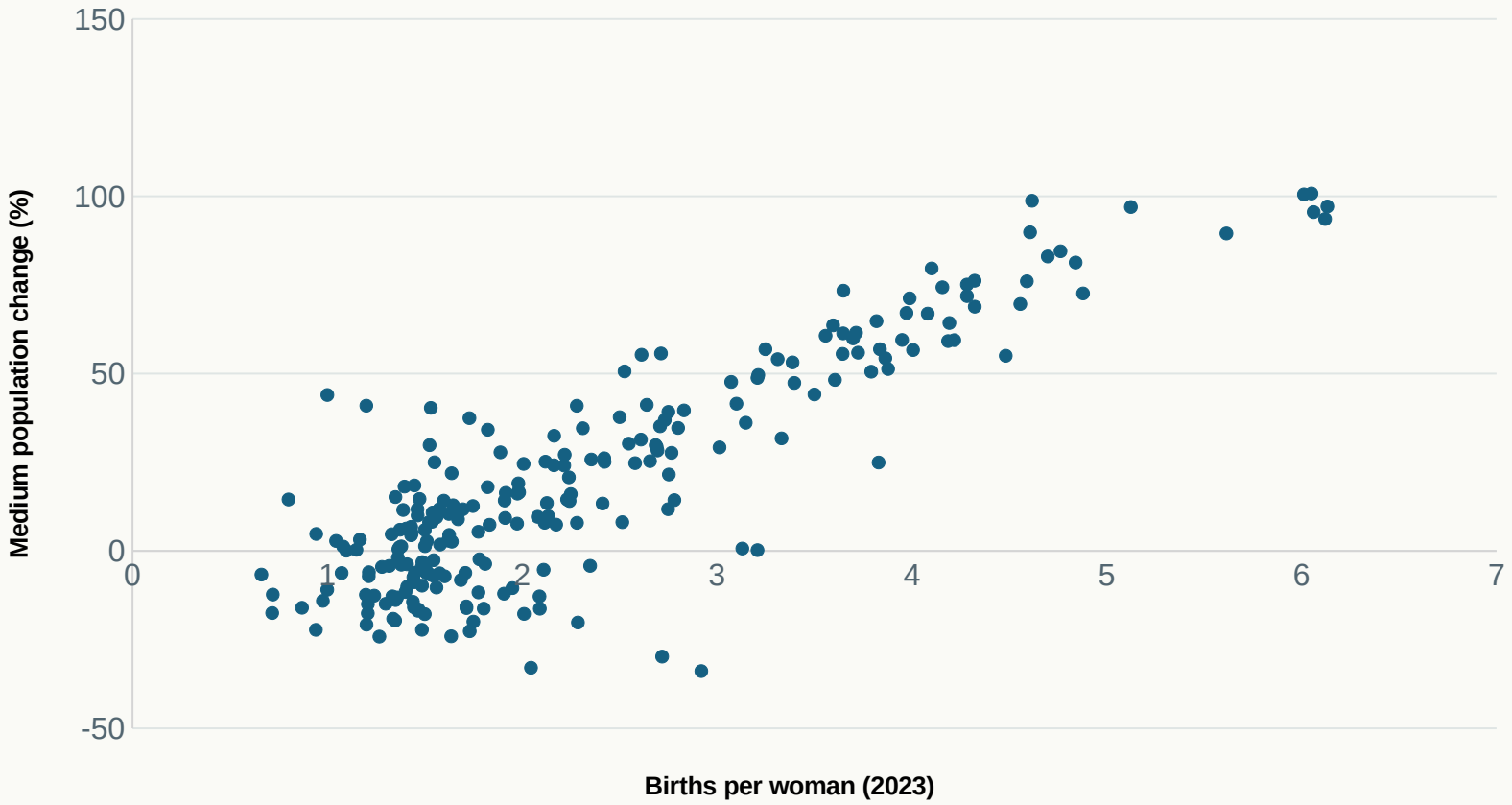
Population momentum and uncertainty to 2050

237 countries and areas
UN World Population Prospects 2024

A scenario analysis with a Türkiye case study

Low fertility and growth coexist in the Medium projection

Fertility in 2023 and population change from 1 Jan 2024 to 1 Jan 2050



64 / 130

low-fertility places grow in the Medium projection

58%

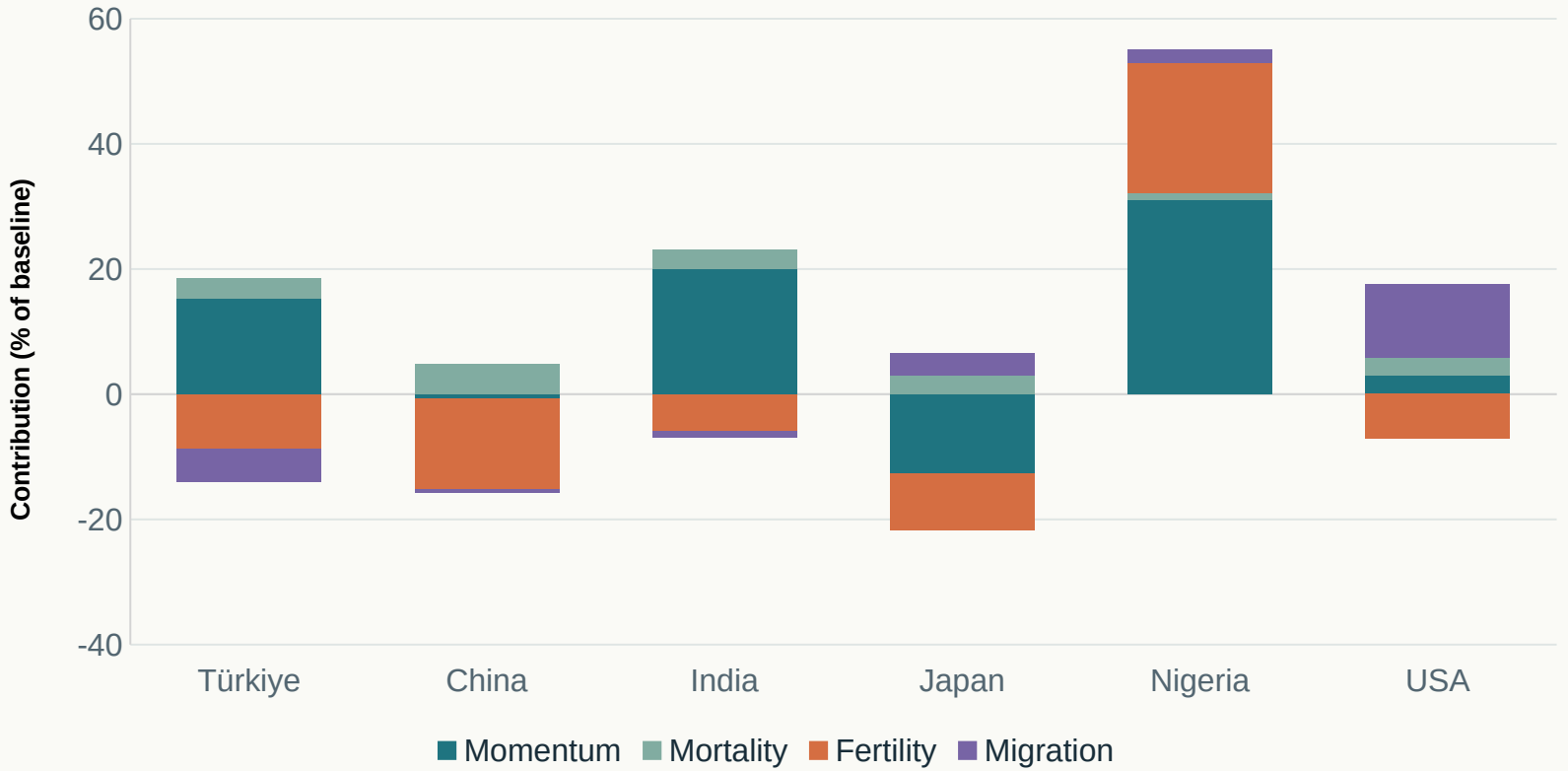
of the low-fertility group's 2023 population lives there

2.1 is an approximate threshold.
One point = one country or area.

UN World Population Prospects 2024. Estimates through 2023. Projections from 2024.

Scenario contrasts separate four contributions

Change to 1 Jan 2050, expressed as a percentage of the 1 Jan 2024 population



A fixed scenario sequence

Momentum
+ projected mortality
+ projected fertility
+ projected migration

The four stock differences add to Medium change.

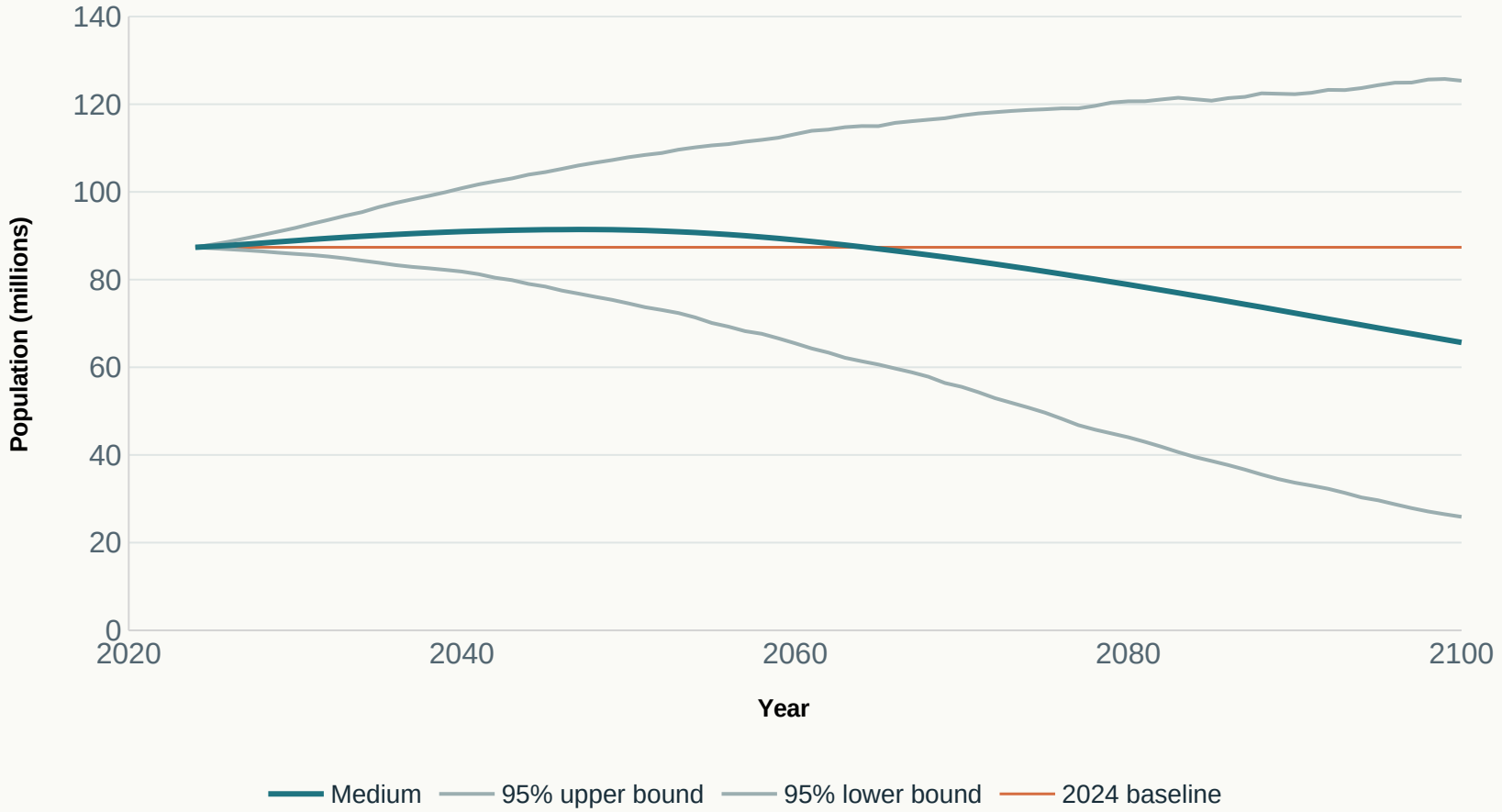
Replacement means NRR = 1.
The fertility rate varies
by country and year.

Contributions depend on the scenario order. They are conditional model contrasts, not causal policy effects.

UN World Population Prospects 2024. Estimates through 2023. Projections from 2024.

Türkiye grows to 2050 in Medium, with a wide interval

Population on 1 January, millions. The 2024 revision does not represent current observed population.



91.31 million

Medium population in 2050

74.57–107.92

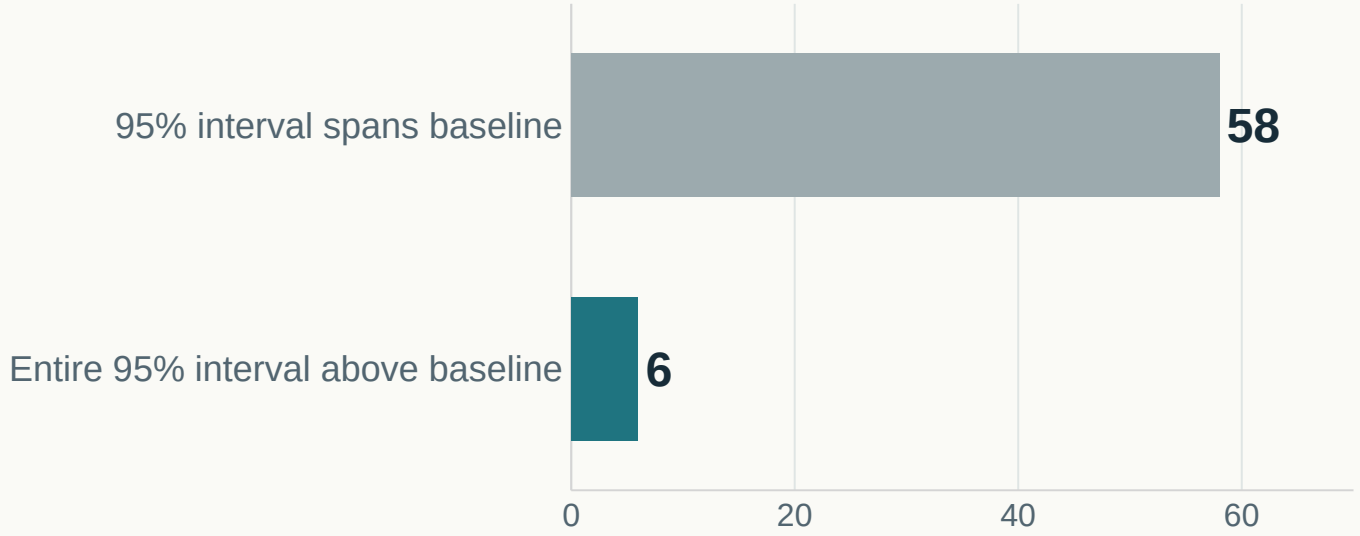
UN 95% prediction interval
The baseline is 87.37 million.

Positive momentum can coexist with later decline.

UN World Population Prospects 2024. Estimates through 2023. Projections from 2024.

Most Medium growth cases remain uncertain

Among the 64 low-fertility countries and areas that grow in the Medium projection



Sensitivity checks

39 / 81 grow with population ≥1 million
66 / 130 grow under NRR < 1
28 / 130 grow by 2100 in the original cohort

What the intervals can tell us

Each interval is marginal for one country.
They do not give a joint interval for the count.
Scenario effects depend on the chosen path.

Ethical interpretation

Disclose model assumptions and uncertainty.
Respect reproductive choices and migrants.
Country counts and population weights differ.

UN World Population Prospects 2024. Estimates through 2023. Projections from 2024.

Sources and reproducible analysis

Data and methods

UN DESA, Population Division (2024)

World Population Prospects 2024. Medium, OtherVariants and competition supplement. CC BY 3.0 IGO.

UN DESA (2024), Methodology, Table II.4, pp. 55–56

Scenario assumptions and the sequential decomposition of projected population change.

UN DESA, Demographic Components of Future Population Growth

Methodological background and reference to Andreev, Kantorová and Bongaarts (2013).

American Statistical Association (2022)

Ethical Guidelines for Statistical Practice.

Analysis package and acknowledgement

Independent scripts, raw-data hashes and validation results are included. Prior EDA: idiltuzkaya/asa_dataquest.

AI assisted the coding, analysis checks and presentation. Team members should review and disclose their contributions.